

Seaborn Plot Types Comparison Table

 Plot Type	 Primary Use	 Key Parameters	 Example (with tips dataset)
scatterplot()	Show relationship between 2 numeric vars	x, y, hue, style, size, data	<code>sns.scatterplot(x="total_bill", y="tip", data=tips)</code>
lineplot()	Show trends or changes over ordered var	x, y, hue, data, marker, style	<code>sns.lineplot(x="size", y="tip", data=tips.groupby("size").mean().reset_index())</code>
barplot()	Show average value of a category	x, y, hue, data, ci, estimator	<code>sns.barplot(x="day", y="total_bill", data=tips)</code>
countplot()	Count observations in categories	x, hue, data	<code>sns.countplot(x="day", data=tips)</code>
boxplot()	Show distribution (quartiles, outliers)	x, y, hue, data	<code>sns.boxplot(x="day", y="total_bill", data=tips)</code>

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violinplot()	Boxplot + KDE (smooth distribution)	x, y, hue, data, split	sns.violinplot(x="day", y="total_bill", data=tips)
stripplot()	Scatter of all points with jitter	x, y, data, jitter, hue	sns.stripplot(x="day", y="total_bill", data=tips, jitter=True)
swarmplot()	Like strip, but avoids overlap	x, y, data, hue	sns.swarmplot(x="day", y="total_bill", data=tips)
histplot()	Distribution of single variable	x, data, bins, kde	sns.histplot(x="total_bill", data=tips, bins=20)
kdeplot()	Smooth density estimation (KDE)	x, data, fill, bw_adjust	sns.kdeplot(x="total_bill", data=tips, fill=True)
lmpplot()	Linear regression line + scatter	x, y, data, hue, col, row	sns.lmpplot(x="total_bill", y="tip", data=tips)

 Plot Type	 Primary Use	 Key Parameters	 Example (with tips dataset)
relplot()	General function for scatter/line grid	x, y, data, kind, col, row, hue	sns.relplot(x="total_bill", y="tip", kind="scatter", data=tips, col="day")
heatmap()	Show values in matrix (correlation etc.)	data, annot, cmap	sns.heatmap(tips.corr(numeric_only=True), annot=True, cmap="coolwarm")
pairplot()	Pairwise scatterplots with histograms	data, hue, kind, diag_kind	sns.pairplot(tips, hue="sex")
catplot()	Wrapper for categorical plots (flexible)	x, y, data, kind, hue, col, row	sns.catplot(x="day", y="tip", kind="box", data=tips)

Parameter	Purpose	Accepted Types	Commonly Used In
x	Variable for x-axis	Numeric / Categorical	All plots
y	Variable for y-axis	Numeric / Categorical	All plots
hue	Group data by color	Categorical	scatterplot, boxplot, violinplot
size	Vary marker size by value	Numeric	scatterplot, relplot
style	Vary marker shape	Categorical	scatterplot, relplot
col	Create columns in facet grid	Categorical	relplot, catplot, Implot
row	Create rows in facet grid	Categorical	relplot, catplot, Implot

Parameter	Purpose	Accepted Types	Commonly Used In
kind	Type of plot (scatter, line, box, etc.)	String	relplot, catplot
data	Input DataFrame	DataFrame	All plots
jitter	Add noise to separate overlapping points	Boolean / Float	stripplot, swarmplot
ci	Show confidence intervals	Integer / 'sd' / None	barplot, pointplot
estimator	Function to aggregate data (e.g. mean, sum)	Function	barplot, pointplot
palette	Set color palette	String / List	All plots
markers	Custom marker styles	Boolean / List	scatterplot, relplot
dodge	Separate hue levels in plots	Boolean	boxplot, violinplot

A kernel density estimate (KDE) plot is a method for visualizing the distribution of observations in a dataset, analogous to a histogram.